دانشگاه زنجان

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(COURSE DESCRIPTION)

DATE: 2026/09/10
INDEX: 27016

University of Zanjan - Zanjan
Faculty of Electrical and Computer Engineering
Master's Degree
Computer Engineering - Software

Remarks:
According to the Academic Instruction of University, in the academic system, each theoretical credit is presented in 16 or 17 hours, each practical credit in 32 or 34 hours,each of the workshop credit in 48 or 51 (except Professional Certificates) and each internship or training credit in 64 to 120 hours.

There are some different type of the courses passing by a student at this university, as listed below:

1. Basic
2. Core / Major
3. Specialized
4. Elective / Optional
5. General
6. Internship / Workshop
7. Project / Thesis

Each one of these types contain some courses to be passed, according to the educational regulations passed by the Ministry of Science, Research and Technology or Ministry of Health & Medical Education of Iran.

The grading system in this university is from 0 to 20. The minimum passing grade for a course leading to an Associate's Degree or a Bachelor's Degree is 10, for a course leading to a Master's Degree and Medical Sciences is 12 and for a course leading to a PhD Degree is 14.

Note: This document has been prepared at **Zahra Koosha's** request in the **Course Description** Platform. Scan the QR-code below to certify this document.

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Course	Credits	Description
Special Topics in Databases	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** The objective of this course is to provide students with an in-depth understanding of advanced and emerging topics in database systems. Students will explore modern database architectures, query processing and optimization techniques, transaction management, distributed and parallel databases, big data technologies, NoSQL systems, data warehousing, data mining, and cloud-based data management. The course aims to develop the ability to design, analyze, and implement advanced database solutions, and to conduct research in database-related areas. **Syllabus :** Advanced database models and architectures. Query processing and optimization. Advanced indexing and hashing techniques. Transaction management and concurrency control. Recovery and fault tolerance. Distributed database systems. Parallel database systems. Big data storage and processing. NoSQL databases. NewSQL databases. Data warehousing and OLAP. Data mining and knowledge discovery. Data streams and continuous queries. Spatial and temporal databases. Multimedia databases. Web databases and information retrieval. Cloud data management. Database security and privacy. Self-tuning and autonomous databases. Emerging trends in database research.
Advanced Topics in Natural Language Processing	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** This course aims to provide students with an in-depth understanding of advanced techniques and recent trends in natural language processing (NLP). Students will learn to analyze and implement state-of-the-art models for various NLP tasks, including deep learning architectures, transfer learning, and large language models. The course emphasizes both theoretical foundations and practical applications, enabling students to conduct research and develop innovative NLP solutions. **Syllabus :** Introduction to advanced NLP and recent trends. Word embeddings and contextual representations. Deep learning for NLP: RNNs, CNNs, and attention mechanisms. Transformer architectures and BERT. Pre-trained language models and transfer learning. Fine-tuning and prompt-based learning. Sequence labeling and structured prediction. Syntactic parsing and semantic analysis. Coreference resolution and discourse analysis. Machine translation and multilingual NLP. Question answering and reading comprehension. Text summarization and generation. Dialogue systems and conversational AI. Information extraction and knowledge graphs. Sentiment analysis and opinion mining. Ethical considerations and bias in NLP. Evaluation metrics and benchmarking. Recent advances and future directions.

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Advanced Computer Networks	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** **Syllabus :** Following the prerequisite course of data communication, this course aims to give more advanced topics in the area. Beginning with data link layer, this course presents multiple access control protocols such as CSMA/CD and CSMA/CA along with their performances. The basics of local area networks (LANs) are given and related topics such as physical address, switches, virtual LANs, and MPLS are introduced. Wireless and mobile access networks are presented and different technologies in this area such as wireless local area networks (WLANs) and cellular networks are discussed. The course then proceeds with required topics in network security such as cryptography, digital signature, secure TCP (SSL), IPSec and tunneling. Then, the structure and performance of WLANs in terms of security are investigated. The course ends with advanced topics in multimedia networking applications.
Systems Analysis and Big Data	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** The course aims to equip students with advanced knowledge and skills in analyzing large-scale systems and big data. Students will learn to design and implement scalable data processing pipelines, apply statistical and machine learning techniques to extract insights from massive datasets, and understand the challenges of distributed computing, data storage, and real-time analytics. The course also covers ethical and privacy considerations in big data, preparing students for research and professional roles in data-intensive software engineering. **Syllabus :** Introduction to big data: characteristics, challenges, and opportunities. Big data ecosystem: Hadoop, Spark, and NoSQL databases. Distributed file systems: HDFS and object storage. MapReduce paradigm and beyond. Data stream processing: Storm, Flink, and Kafka. Scalable data storage: columnar stores, key-value stores, and graph databases. Data preprocessing and cleaning for big data. Statistical analysis of large datasets: sampling, estimation, and hypothesis testing. Machine learning at scale: distributed algorithms for clustering, classification, and regression. Dimensionality reduction and feature engineering for big data. Graph analytics and social network analysis. Real-time analytics and complex event processing. Cloud computing for big data: AWS, Azure, and GCP. Performance tuning and optimization of big data systems. Data visualization for big data. Security, privacy, and ethical issues in big data. Case studies and applications in software engineering.

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Advanced Network Security	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** This course delves into advanced topics in network security, equipping students with the knowledge and skills required to design, implement, and manage robust security measures for computer networks. It covers cutting-edge techniques, technologies, and strategies to safeguard data, systems, and communications. **Syllabus :** Understand advanced network threats, vulnerabilities, and attack techniques. Design and implement secure network architectures and protocols. Evaluate and select appropriate security tools and technologies. Analyze network traffic for anomalies and potential threats. Develop incident response plans and strategies. Secure wireless and mobile networks effectively. Stay updated on emerging trends and threats in network security.
Special Topics - Deep Learning (PhD)	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** The course aims to provide an in-depth understanding of deep learning theory, architectures, and applications. Students will learn to design, implement, and evaluate deep neural networks for various tasks. The course emphasizes both foundational principles and recent advances, preparing students for research and practical development in deep learning. **Syllabus :** Introduction to deep learning and historical context. Mathematical foundations: linear algebra, probability, and calculus. Machine learning basics: regression, classification, and regularization. Neural networks: perceptrons, multilayer networks, and backpropagation. Optimization for deep learning: SGD, Adam, and second-order methods. Regularization techniques: dropout, batch normalization, and early stopping. Convolutional neural networks: architectures and operations. CNN architectures: LeNet, AlexNet, VGG, ResNet, and Inception. Recurrent neural networks: LSTM, GRU, and bidirectional RNNs. Sequence modeling: language models, machine translation, and attention mechanisms. Transformers: self-attention, BERT, and GPT. Generative models: autoencoders, variational autoencoders, and GANs. Deep reinforcement learning: DQN, policy gradients, and actor-critic methods. Unsupervised and self-supervised learning: clustering, contrastive learning, and pretraining. Deep learning for computer vision: object detection, segmentation, and image generation. Deep learning for natural language processing: text classification, sentiment analysis, and question answering. Graph neural networks: basics and applications. Deep learning frameworks: TensorFlow and PyTorch. Hardware and distributed training: GPUs, TPUs, and parallelization. Ethical considerations and fairness in deep learning. Recent advances and research trends in deep learning.

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Seminar	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** To familiarize students with how to gather theoretical and research materials using the world's current common methods in connection with a specific topic and present the prepared material along with analysis in an oral presentation. **Syllabus :** Literature review on one of the topics related to the selection, identification, and properties of engineering materials, including preparing a list of the latest scientific articles in the chosen field using the latest methods of searching scientific sources and references, collecting articles by conducting theoretical studies and critiquing the work done, summarizing them and drawing final conclusions - no experiments are carried out in the seminar and only the review and analysis of materials conducted by other researchers is done. The topic of the seminar should be different from the title of the master's project, but it can include part of the initial studies of the project. The seminar, after being typed and edited, is presented in a pre-determined session with the presence of the supervisor professor, the head of the department, and the students of the relevant department.
Data Mining	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** The goal of this course is to familiarize students with the concepts and challenges involved in exploring large volumes of data and discovering underlying patterns within them. The course begins with an introduction to data preprocessing and data warehousing concepts, followed by a discussion of key topics in the field of data mining. **Syllabus :** Introduction to data mining and its main models. Data preprocessing methods. Introduction to data warehousing, data cubes, and analytical reporting. MapReduce model. MapReduce algorithms. Complexity theory of the MapReduce model. Introduction to related technology stacks such as Spark, Hadoop, etc. Mining itemsets in large volumes. Finding similar itemsets and locality-sensitive hashing functions. Finding frequent itemsets, A-Priori algorithm, PCY algorithm, SON algorithm, and Toivonen algorithm. Data stream mining. Data stream sampling. Data stream filtering and Bloom filters. Data stream algorithms. Analysis of data networks. Structural properties of networks. PageRank and random walk. Community detection algorithms. Motif counting algorithms in graphs. Graph sampling algorithms. Data clustering algorithms. Hierarchical algorithms. K-Means algorithm. CURE algorithm. Data stream clustering. Dimensionality reduction algorithms. SVD and PCA algorithms. CUR algorithm. Machine learning algorithms for big data. Concept of machine learning and regression algorithms. Neural networks, introduction to deep neural networks. SVM algorithm. Data mining in industry. Recommender systems. Web advertising and the Adwords problem. Matching markets. Pricing problem. Healthcare data mining.

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Computational Intelligence	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** The main objective of this course is to introduce students to the fundamental concepts and techniques of computational intelligence, including neural networks, fuzzy systems, evolutionary algorithms, and swarm intelligence. Students will learn how these paradigms can be used to solve complex real-world problems that are difficult to address with traditional methods. The course aims to develop the ability to design, implement, and evaluate computational intelligence systems, and to apply them in various domains such as pattern recognition, control, optimization, and decision making. **Syllabus :** Introduction to Computational Intelligence and Soft Computing. Artificial Neural Networks: Perceptron, Multilayer Perceptrons, Backpropagation. Radial Basis Function Networks. Support Vector Machines. Unsupervised Learning: Hebbian Learning, Competitive Learning, Self-Organizing Maps. Adaptive Resonance Theory. Fuzzy Sets and Fuzzy Logic. Fuzzy Inference Systems: Mamdani and Sugeno. Fuzzy Control. Evolutionary Computation: Genetic Algorithms, Genetic Programming, Evolution Strategies, Evolutionary Programming. Swarm Intelligence: Ant Colony Optimization, Particle Swarm Optimization. Artificial Immune Systems. Hybrid Systems: Neuro-Fuzzy Systems, Evolutionary Neural Networks, Fuzzy Evolutionary Systems. Deep Learning: Convolutional Neural Networks, Recurrent Neural Networks. Applications of Computational Intelligence in Pattern Recognition, Image Processing, Robotics, and Data Mining. Ethical and Social Implications of Computational Intelligence.
Thesis	6	**(Duration: 32 hours Practical & 80 hours Theoretical)** **(Type of Course: Thesis)** The dissertation is of great importance because of scientific and research activities and in fact is a criterion for measuring the quality and quantity of research work of scholars and the level of science and research. The main purpose of this dissertation is to acquaint students with the method of scientific research. Better and more accurate knowledge of the scientific foundations of specialized topics, mastery of how to collect, organize and analyze scientific information and use of valid and new sources, is the most important achievement of research activities in the form of papers. **Syllabus :** The dissertation is equivalent to 4 or 6 credits. In the dissertation section, the students of the course study a subject in medical, industrial, research, software and environmental trends. And acquire the necessary skills appropriate to the specialized orientations. It is necessary to define the topic of the dissertation as much as possible to meet the needs of the country and with priority on major national, regional and local issues.

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